frequencies. Also, as these inverters rely on the input power waveform for a clock timer, they will not work properly. Talking of advantages, modified sinewaves have very low harmonic distortion compared to the other waveforms. These inverters run inductive loads like microwave ovens and motors correctly; reduce audible and electrical noise in audio amplifiers, TVs, fans, fluorescent lights, fax machines and game consoles; and prevent crashes in computers, and glitches and noise in monitors.

Techniques to improve power conversion

Pulse-width modulation technique is used to power applications like electric tubelights, kitchen appliances, power tools, TV sets, radios, computers and many more electronics gadgets. This technique provides much cleaner AC power than modified sinewave and squarewave inverters. However, there are two major requirements: keeping inverter boxes watertight and using components that can withstand high temperatures. It is necessary to develop a system that can detect failure of any micro inverter and isolate it immediately.

Some of the techniques to improve power conversion efficiency of inverters are listed below:

1. To achieve zero-voltage switching, use an auxiliary inductor and two auxiliary switches in a single-phase inverter with neutral-point clamped topology and fixed-frequency PWM control. This can help to improve inverter efficiency.

2. To reduce total harmonic distortion as well as to control output voltage, use sinusoidal pulse-width modulation technique.

3. To improve the voltage, current or capacity level of power-conversion systems, connect multiple standardised modules in series or parallel at the input/output.

4. It is possible to protect inverters from inrush and over current automatically by adding clamp circuits to the resonant capacitance in parallel. High power-conversion efficiency can be achieved by regenerating the clamp current to the input voltage source.

5. To achieve near-zero common-mode voltage generation for a three-phase inverter, neutral-point diode-clamping is used. This solves desynchronisation issue of the balanced inverter.

6. A current-source-type four-switch inverter based on switch multiplexing technique can be used to reduce the input low-frequency current ripple as well as DC capacitor requirement. It is proved to be input-ripple-free and more efficient, and its size is much smaller because the DC capacitor size can be reduced significantly.

Way forward

We are moving towards solar power but it is still a developing technology with less than one per cent of consumers having access to it. So inverters are here to stay for at least next 50 years, strengthening the case for their improvement to make these more efficient. EFY